

Name: _____

Date: _____

DIRECTIONS

This Independent Practice is for Lesson 4.11.01.

Show your work in the box. Write your final answer on the line.

Read each problem carefully. Use a model or picture if it helps.

If you get stuck, re-read the question and look at any example you already worked through in the lesson.

1 Build the rule

Area of a rectangle = length $\times$ width. Both sides must be in the SAME unit.

(a) $6 \text{ in} \times 4 \text{ in} = \underline{\hspace{1cm}}$ sq in

(b) $9 \text{ cm} \times 7 \text{ cm} = \underline{\hspace{1cm}}$ sq cm

(c) $12 \text{ ft} \times 5 \text{ ft} = \underline{\hspace{1cm}}$ sq ft

MULTIPLY L $\times$ W

(a) _____ (b) _____ (c) _____

2 Squares — both sides equal

A square has equal sides. Area = side $\times$ side.

(a) A 5-in square tile. Area = ____ sq in.

(b) An 8-ft square garden. Area = ____ sq ft.

(c) Why does a square need only ONE side measurement?

SIDE $\times$ SIDE; EXPLAIN

(a) ____ (b) ____ (c) ____

3 Mixed units — convert first

Both sides must be in the SAME unit before you multiply.

(a) $2 \text{ ft} \times 8 \text{ in} =$ ____ sq in (convert $2 \text{ ft} = 24 \text{ in}$ first)

(b) $1 \text{ yd} \times 22 \text{ cm} =$ ____ sq cm ($1 \text{ yd} \approx 91 \text{ cm}$)

(c) $1 \text{ yd } 1 \text{ ft} \times 5 \text{ ft} =$ ____ sq ft (convert $1 \text{ yd } 1 \text{ ft} = 4 \text{ ft}$ first)

CONVERT; THEN MULTIPLY

(a) ____ (b) ____ (c) ____

4 Same area, different shape

Find rectangles with area exactly 24 sq cm.

- (a) Give one rectangle (length and width) with area 24 sq cm.
- (b) Give a DIFFERENT rectangle (different shape) with area 24 sq cm.
- (c) How many different rectangles with WHOLE-NUMBER sides have area 24 sq cm?

FIND FACTOR PAIRS OF 24

(a) ____ × ____ (b) ____ × ____ (c) ____

5 Find and fix

A student wrote: "A rectangle 3 ft long and 9 in wide has area 27 sq feet."

- (a) What is wrong with their answer?
- (b) Find the correct area in square inches.

FIND THE UNIT ERROR

(a) ____ (b) ____ sq in

6 Word problem — Maya's garden

Maya is planting a rectangular garden that is 8 ft long and 5 ft wide.

(a) What is the area of the garden in square feet?

(b) Maya needs 1 cup of seed for every 10 sq ft. How many cups of seed does she need? (Round up.)

AREA; THEN DIVIDE & ROUND UP

(a) ____ sq ft (b) ____ cups